

MedVision AI

Medical Image AI Pipeline — Brain MRI Classification and Segmentation

Learning Resources and Contributor Onboarding

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Abstract

This document provides a curated list of courses (Coursera, Pluralsight, A Cloud Guru), training repositories, and reference books to help contributors ramp up on the technologies used in **MedVision AI**. Resources are selected for quality, relevance, and practical applicability.

Contents

Prerequisites

Before starting the courses below, ensure you are comfortable with: Python, linear algebra (matrices, eigenvalues), calculus (gradients), basic statistics.

Coursera Courses

Course	Link	Why relevant
AI for Medicine Specialization	link	Medical image analysis, diagnostic AI, treatment effect estimation — directly applicable.
Deep Learning Specialization	link	CNN architecture theory, optimisation, regularisation.
Practical Data Science on AWS	link	MLOps, experiment tracking, model deployment patterns.

Table 1: Coursera

Pluralsight Courses

Course	Link	Why relevant
PyTorch: Getting Started	link	Tensors, autograd, DataLoader — foundations for this project.
MLflow for ML Experiment Tracking	link	Direct tool match.

Table 2: Pluralsight

A Cloud Guru Courses

Course	Link	Why relevant
Machine Learning on AWS	link	SageMaker patterns for scalable model training.

Table 3: A Cloud Guru

GitHub Training Repositories

- [Project-MONAI/MONAI](#) — Medical image deep learning framework; transforms, datasets, metrics.
- [mlflow/mlflow](#) — Experiment tracking tool used in this project.
- [iterative/dvc](#) — Data version control; pipeline stages, remote storage.
- [pytorch/vision](#) — DenseNet, ResNet, Swin-V2 implementations.

Reference Books

- *Deep Learning for Medical Image Analysis* — Zhou et al.. Segmentation, classification, and detection in radiology.
- *Programming PyTorch for Deep Learning* — Ian Pointer. Practical PyTorch patterns for transfer learning.

Suggested Learning Path

1. Complete prerequisites (1–2 weeks).
2. Work through the Coursera specialisation most relevant to the repo's domain (4–8 weeks).
3. Clone the repository, read `README.md` and `INFRASTRUCTURE.md`.
4. Run the service locally following the **Local Run** section of `INFRASTRUCTURE.md`.
5. Study the Pluralsight courses for the specific tools used (2–4 weeks, parallel).
6. Contribute a small fix or documentation improvement as a first PR.
7. Study the GitHub training repositories for advanced patterns.

Free Resources

- **Python Official Tutorial** — free, authoritative.
- **Kubernetes Interactive Tutorials** — browser-based k8s labs.
- **FastAPI Tutorial** — official, comprehensive.
- **PyTorch Tutorials** — official, covers all levels.
- **DeepLearning.AI Short Courses** — free, recent AI topics.